Chemical diversity in European beech

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Context

Specialised metabolites play an essential role in plants' interactions with their surroundings, such as providing defences against biotic stresses, enabling communication within the community, or conferring resistance to abiotic stresses. In beech (*Fagus sylvatica*), a few studies have examined individual chemical profiles (Cadahía et al. 2015) or compound variation in response to a stress factor (Eisenring et al. 2024); fewer have examined natural variation in the metabolome across the species' distribution. Using high-throughput metabolomics, we aim to study the natural variation of specialised metabolism across the species' range and investigate whether genetic differences among populations shape variation in metabolites involved in crucial defences.

Background

In 2025, we sampled, extracted DNA, and sequenced whole genomes from 96 beech seedlings to study genetic variation within and among seed families across 12 beech populations in Europe. These seedlings are part of a macrogarden experimental network to understand survival and performance of beech provenances in the changing European climate (Csilléry et al., 2026; *MyGardenOfTrees*, n.d.). We then used a subset of these samples to develop and test a high-throughput metabolomic extraction and measurement protocol, including testing gradient length (6 min vs 10 min, Fig. 1), which determines the time needed to analyse extracts and thus the number of samples that can be analysed per day. The detailed protocols for this analysis are provided as separate documents.

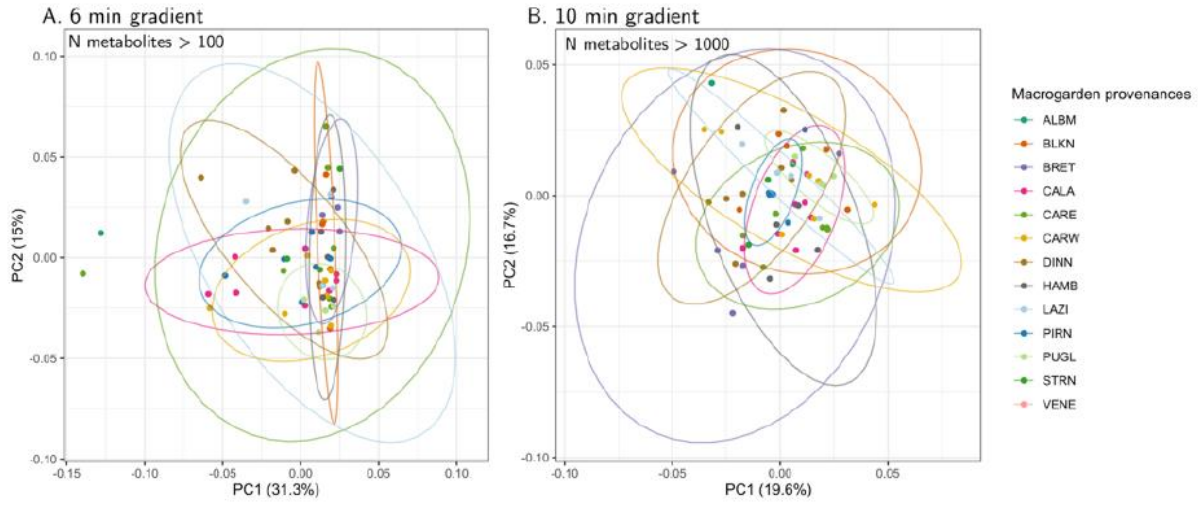


Figure 1. Results of PCA across metabolomic profiles of genetically diverse beech seedlings analysed by LC-MS using a 6-minute or 10-minute chromatographic gradient.

Project

In this project, you will use raw LC-MS data to do the following tasks.

- Evaluate the quality of acquisition of raw data (note: there are known errors and problems in the batch test; it should be a good dataset to test whether raw data checkups are performed, even though mzmine will notify that there are some problems)
- Test whether a shorter gradient captures and represents a similar chemical profile to a longer gradient (note: a shorter gradient would allow us to pass more samples within one batch; however, we need to evaluate the trade-off between information and the number of samples acquired – in this part, PLS-DA and machine learning tests could be implemented and then compared to the results)
- Identify whether there are specific compounds that differentiate the two species included in the batch test (data analysis expected: as the design is unbalanced, it could also introduce some statistical concepts)
- Propose recommendations for data acquisition or design of study

Please also note that some tubes are left from the batch test with powder (stored at -20C); these could be used to perform the protocol and then create another batch test.

Cadahía, Estrella, Brígida Fernández de Simón, Ismael Aranda, Miriam Sanz, David Sánchez-Gómez, and Ernani Pinto. 2015. "Non-Targeted Metabolomic Profile of Fagus Sylvatica L. Leaves Using Liquid Chromatography with Mass Spectrometry and Gas Chromatography with Mass Spectrometry: Fagus SylvaticaLeaf Metabolomic Profile." *Phytochemical Analysis: PCA* 26 (2): 171–82.

Eisenring, Michael, Arthur Gessler, Esther R. Frei, et al. 2024. "Legacy Effects of Premature Defoliation in Response to an Extreme Drought Event Modulate Phytochemical Profiles with Subtle Consequences for Leaf Herbivory in European Beech." *The New Phytologist* 242 (6): 2495–509.